1. A life-cycle assessment looks at the potential environmental impact at each stage of the life of a product.

Cat food is sold in plastic packets or metal cans.



The table shows information about these two containers.

	Plastic packet	Metal can
Raw Materials	Crude oil	Aluminium ore
Manufacture	Fractional distillation Cracking Polymerisation	Aluminium ore is mined Aluminium is extracted by electrolysis
Using the product	Usually single use	Usually single use but easily repurposed or upcycled
Disposal of the product	Harder to recycle Not biodegradable so takes up space in landfill 90% energy saved by recycling	Easier to recycle Not biodegradable so takes up space in landfill 95% energy saved by recycling

Evaluate which type of container has the smallest environmental impact over its lifetime.
Use the information in the table, and your own scientific knowledge.

[6]

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Mark scheme

Question			Answer/Indicative content	Marks	Guidance
1			**Level 3 (5–6 marks)** Analyses the information to give a clear and detailed discussion of the environmental impacts of each container over its lifetime. **AND** Clear evaluation, that supports information from the table, of which container has the smallest environmental impact. *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3–4 marks)** Analyses the information to discuss the possible environmental impacts of each container, but there is limited detail. **AND** Clear evaluation of which container has the smallest environmental impact. *There is a line of reasoning presented with some structure. The information presented is relevant and supported by some evidence.* **Level 1 (1–2 marks)** Analyses the information to give a basic discussion of some of the environmental impacts of each container. *There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.* **0 mark** *No response or no response worthy of credit.*	6 (6 × AO 3.2a)	**AO3.1b Analyse ideas and information to evaluate** **Raw Materials & Manufacture** • the raw materials used to make plastic packets come from crude oil, which is a non-renewable resource • obtaining crude oil from the ground, fractional distillation, cracking and polymerisation requires a lot of energy • mining of aluminium uses up limited resources and damages the environment • extraction of aluminium by electrolysis uses a lot of energy • the energy in both manufacturing processes often comes from burning fossil fuels so causes the release of greenhouse gases **Use & disposal of the product** • both are usually single use which could increase the amount of waste sent to landfill however metal cans are easier to repurpose/upcycle • plastic packets are non-biodegradable so will stay in the environment for a long time if sent to landfill • plastic packets can be recycled, though it is harder to do so, and recycling uses only 10% of the energy needed to make the plastic from crude oil • metal cans are non-biodegradable so will stay in the environment for a long time if sent to landfill • metal cans can be more easily recycled, and recycling uses only 5% of the energy needed to extract aluminium from aluminium ore • recycling reduces the use of valuable raw materials
			Total	**6**	

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